Category	Description	Probability (1-5)	Impact (1-5)	Risk Level	Owner	Response Strategy	Status
Technical	Garage sensors/hardware may report inaccurate occupancy data.	3	3	Medium	Ironhide	Mitigate - Provide option to manually override in case of discrepancies	Monitoring
Technical	Third party mapping API may have downtime or limits.	2	4	Medium	Soundwave	Mitigate -Provide fallback caching or directions if API goes down	Monitoring
Technical	Payment gateway integration may fail or reject valid payment due to provider side changes.	2	4	Medium	Soundwave	Mitigate - Include retry logic and monitor provider's status page	Monitoring
Technical	Synchronization in real time between the driver application and the operator dashboard could be slow or not work under heavy usage.	2	3	Medium	Ironhide	Mitigate - Perform load testing prior to Final Delivery milestone	Monitoring
Technical	Interference from GPS/structure (concrete floor/underground floor) can result in inaccurate "last 50 feet" navigation.	3	2	Medium	Soundwave	Accept - An acceptable limitation; supplement by putting signs up at garage entrance	Open
Technical	The addition of multiple garages simultaneously can result in problems in database performance not seen during small scale testing.	3	3	Medium	Ironhide	Mitigate - Conduct load testing early in process and scale horizontally	Monitoring
Schedule	Long Infrastructure & Integrations phase might put off every dependent feature.	3	4	High	Charlotte Cheng	Mitigate - Closely monitor progress towards milestone; include scheduling buffer	Closed
Schedule	Third-party API onboarding (e.g., mapping or payments) might take more time than planned.	2	3	Medium	Ironhide	Mitigate - Start onboarding early; find backup providers	Closed
Schedule	Driver App and Operator Dashboard teams might become misaligned if one of them delivers their part late.	3	3	Medium	Charlotte Cheng	Mitigate - Enforce contract of API between teams early	Open
Schedule	Testing & QA phase might be shortened because previous phases would overrun.	3	3	Medium	Peter Cullen	Mitigate - Protect QA phase timeline; escalate early in case other milestones fall behind	Open
Schedule	Bugs associated with Integration appear during Testing & QA but were not discovered when testing separately.	4	3	High	Peter Cullen	Accept - Already included in QA phase buffer	Open
Financial	Fees charged by the payment processor may be higher than anticipated.	3	2	Medium	Charlotte Cheng	Accept - Build fee into cost model	Open
Financial	Fees for mapping/navigation API may increase more than expected due to increased traffic.	3	2	Medium	Charlotte Cheng	Accept - Monitor usage; set up budget threshold alert	Open
Financial	The garage operators might not want to pay for premium dashboards features.	3	3	Medium	Optimus Prime	Mitigate - Validate price tiers by conducting pilot program with operators	Open
Financial	Cost overrun when extending the infrastructure phase if there are problems with integration.	2	3	Medium	Charlotte Cheng	Mitigate - Create budget contingency for Infrastructure Phase	Open
Financial	Unexpected legal fees when conducting legal reviews by the city parking compliance.	2	3	Medium	Charlotte Cheng	Transfer - Engage outside legal counsel as needed by city	Open
People	The main backend developer might decide to walk out of the project midway, delaying the infrastructure module because it is a blockage for all other modules.	2	4	Medium	Charlotte Cheng	Mitigate - Thorough documentation of system; no single point of knowledge	Open
People	There could be misunderstandings among the Driver App and Operator Dashboard development teams, which could lead to duplicated work.	3	2	Medium	Charlotte Cheng	Mitigate - Enforce API contracts early between teams	Open
People	Slow response from the garage operator regarding garage occupancy information/data might delay testing.	3	3	Medium	Starscream	Mitigate - Set data submission deadline in Operator Onboarding Agreement	Open
People	Skill gap within the team with respect to the API of the selected payment service provider might delay implementation.	2	3	Medium	Ironhide	Mitigate - Include developer training period/documentation	Open
People	Communication silos amongst 4 different teams that will run in parallel, considering the scaling effect of communication channels with respect to team size.	4	3	High	Charlotte Cheng	Mitigate - Enforce regular meeting cadence	Open
People	Turnover of the garage operator company.	2	2	Low	Starscream	Accept - Low probability; re-onboard new contact in case of issue	Open
Financial	Jira's free plan caps at 10 users, and the team has 15.	5	1	Low	Charlotte Cheng	Accept - Budgeted in the Licenses line (Jira Standard, about $356)	Open
Schedule	The Jira board drifts from real progress and hides delays.	3	3	Medium	Charlotte Cheng	Mitigate - Update at daily stand-up; check against the Gantt chart weekly	Open
Technical	A release goes out with no staging test or rollback plan.	3	5	High	Wheeljack	Mitigate - Staging environment, automated regression tests, and a rollback plan before release	Open